Hadamard Amplitudes and Zero Spacing in Automorphic L-functions

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Draft: April 2026

Abstract

For an L-function $L(s)$ in the Selberg class with simple zeros $\rho_n = \frac{1}{2} + i\gamma_n$ on the critical line, the *Hadamard amplitude* $A(\gamma_n) = \mathrm{Im}(c_0)^2 + 2\,\mathrm{Re}(c_1)$, extracted from the Laurent expansion $\Lambda'/\Lambda(\rho_n + u) = u^{-1} + c_0 + c_1 u + \cdots$, measures the local spectral weight of each zero. We prove the unconditional lower bound $A(\gamma_n) \geq 2/\Delta_{\min}(\gamma_n)^2$, where $\Delta_{\min} = \min(\gamma_n - \gamma_{n-1}, \gamma_{n+1} - \gamma_n)$ is the minimum spacing to a neighbouring zero, and demonstrate numerically that the Spearman anti-correlation $\rho(A, g_{\min})$ is (i) *universal*: consistently negative across eleven automorphic L-functions of degree 1 through 4 under the canonical $(A_\Lambda, g_{\min}^{\mathrm{GUE}})$ pairing, with $\rho_{\mathrm{S}} = -0.894 \pm 0.014$ (range 0.043), including both self-dual ($\varepsilon = \pm 1$) and non-self-dual ($\varepsilon \in S^1$) families; further verified across twelve L-functions of degree 1 through 6 under mixed metrics, (ii) *height-stable*: the correlation is t-independent for the completed amplitude A_Λ paired with $g_{\min}$ (span 0.16 across eight height bins), and (iii) *GUE-predicted with degree-dependent damping*: under method-matched conditions (± 300 local sum, 20% trim), the GUE ensemble gives $\rho_{\mathrm{S}} = -0.967$, while $\zeta(s)$ gives -0.929 ($\Delta\rho \approx 0.04$, not significant) and GL(2) gives ~ -0.67 ($\Delta\rho \approx 0.24$, $\sim 2\sigma$). The previously reported universal damping $\sim$0.38 is a methodological artefact conflating T-range truncation, sum locality, and genuine arithmetic attenuation. These results establish $A(\gamma)$ as a new invariant linking individual zero strength to local spectral spacing: the anti-correlation sign is degree-independent across the Langlands hierarchy, while its magnitude exhibits degree-dependent arithmetic damping.

1 Introduction

The distribution of nontrivial zeros of L-functions has been a central object of study since Riemann's 1859 memoir. While the vertical distribution (zero density, pair correlation) is well understood through the work of Montgomery [1], Odlyzko [2], and others, the *individual* character of each zero—a measure of how "strongly" a given zero contributes to the analytic structure—has received comparatively little attention.

In this paper we introduce the *Hadamard amplitude* $A(\gamma)$, a nonnegative real number attached to each simple zero $\rho = \frac{1}{2} + i\gamma$ of a completed L-function $\Lambda(s)$. The amplitude is defined via the Laurent expansion of the logarithmic derivative at the zero:

$$\frac{\Lambda'}{\Lambda}(\rho + u) = \frac{1}{u} + c_0 + c_1 u + c_2 u^2 + \cdots, \tag{1}$$

where the coefficients c_n carry arithmetic and spectral information. By the parity theorem (Theorem 5 of [4]), c_0 is purely imaginary and c_1 is purely real for on-critical zeros, so that

$$A(\gamma) \;=\; \mathrm{Im}(c_0)^2 + 2\,c_1 \;=\; B(\gamma)^2 + 2\,H_1(\gamma), \tag{2}$$

where $B = \mathrm{Im}(c_0)$ and $H_1 = c_1 = \mathrm{Re}(c_1)$. The Hadamard product representation yields the explicit decomposition

$$B(\gamma_n) = -\sum_{k \neq n}{}' \frac{1}{\gamma_n - \gamma_k} + \Gamma\text{-terms}, \qquad H_1(\gamma_n) = \sum_{k \neq n}{}' \frac{1}{(\gamma_n - \gamma_k)^2} + \Gamma'\text{-terms}, \tag{3}$$

where the primed sum runs over all nontrivial zeros γ_k of $\Lambda(s)$ (see §2 for precise definitions). The sum B is alternating (cancellative), while H_1 is strictly positive—this asymmetry is the source of the results that follow.

Main contributions.

(i) **Unconditional lower bound** (theorem 3.1): $A(\gamma) \geq 2/\Delta_{\min}(\gamma)^2$, a direct consequence of the positivity of H_1.

(ii) **Degree-independent universality** (section 5): the Spearman anti-correlation $\rho(A, g) < 0$ is verified across twelve L-functions of degree 1 through 5, including Riemann ζ, Dirichlet L-functions, elliptic-curve L-functions, and symmetric powers up to sym^4.

(iii) **GUE mechanism** (section 7): a GUE simulation reproduces 86% of the observed correlation; the residual 14% is explained by the Gamma-factor cross-term.

(iv) **Canonical metric selection** (section 6): $g_{\min}$ is height-stable (Kendall $\tau = -0.29$, $p = 0.40$), while g_{right} decays with height ($\tau = +1.0$, $p = 5 \times 10^{-5}$), identifying $g_{\min}$ as the canonical gap metric.

(v) **Methodology correction** (remark 5.2): the naïve gap $|\rho_{\text{GUE}}| - |\rho_L| \approx 0.38$ conflates three distinct effects: T-range truncation, locality of the Hadamard sum, and genuine arithmetic damping. Under method-matched conditions the residual arithmetic attenuation $\Delta\rho \approx 0.04$ for $d = 1$ at $W = 300$ vanishes exponentially as W grows, achieving full GUE universality; for $d = 2$, $\Delta\rho \sim 0.24$ (T-limited).

Relation to prior work. The amplitude $A(\gamma)$ is the curvature coefficient in the ξ-bundle framework of [3, 4, 5]: the identity $\kappa\delta^2 - 1 = A\delta^2 + O(\delta^3)$ (Corollary 4.2 of [4]) shows that A governs the leading deviation of the curvature $\kappa = |\Lambda'/\Lambda|^2$ from its singular $1/\delta^2$ behaviour near a zero. The Hadamard decomposition $A = B^2 + 2H_1$ (Proposition 11 of [4]) connects A to the global zero distribution, while the present paper develops the consequences for *local* spectral spacing.

2 Preliminaries

2.1 Setting and notation

Let $L(s)$ belong to the Selberg class $\mathcal{S}$: it has an Euler product, analytic continuation, and a functional equation

$$\Lambda(s) = \varepsilon\, \overline{\Lambda(1 - \bar{s})}, \qquad |\varepsilon| = 1, \tag{4}$$

where $\Lambda(s) = \gamma(s)\, L(s)$ is the completed L-function with Gamma factor $\gamma(s)$. We assume all zeros ρ_n on the critical line $\sigma = \sigma_c$ are simple.

We write γ_n for the imaginary parts of the zeros ordered by height: $\gamma_1 < \gamma_2 < \cdots$. The *local density* at height T is $\bar{d}(T) = N(T)/T$, where $N(T)$ counts zeros with $0 < \gamma \leq T$. The *GUE-normalised gap* is

$$g_n^{\text{GUE}} = \frac{\gamma_{n+1} - \gamma_n}{\bar{d}_n^{-1}}, \qquad \bar{d}_n = \frac{2}{\gamma_{n+1} - \gamma_{n-1}}, \tag{5}$$

where $\bar{d}_n$ is the local density estimated from the two-sided gap.

Definition 2.1 (Gap metrics)**.** For the n-th zero γ_n, define:

$$g_{\text{right}}(\gamma_n) = (\gamma_{n+1} - \gamma_n)\, \bar{d}_n, \tag{6}$$

$$g_{\min}(\gamma_n) = \min(\gamma_n - \gamma_{n-1},\ \gamma_{n+1} - \gamma_n)\, \bar{d}_n. \tag{7}$$

2.2 Laurent coefficients and the Hadamard decomposition

At a simple zero $\rho_n = \frac{1}{2} + i\gamma_n$, the logarithmic derivative has a simple pole:

$$\frac{\Lambda'}{\Lambda}(\rho_n + u) = \frac{1}{u} + \sum_{m=0}^{\infty} c_m(\gamma_n)\, u^m.$$

Proposition 2.2 (Parity, [4, Theorem 5])**.** *Under the functional equation* (4) *and the conjugation–compatibility condition* $\Lambda(\bar{s}, \bar{\chi}) = \overline{\Lambda(s, \chi)}$*, the Laurent coefficients satisfy* $c_m = (-1)^{m+1}\, \bar{c}_m$ *for all* $m \geq 0$*. In particular,* $c_0 \in i\mathbb{R}$ *and* $c_1 \in \mathbb{R}$*.*

The Hadamard product for $\Lambda(s)$ gives the explicit representation:

$$c_1(\gamma_n) = \sum_{\substack{k=1 \\ k \neq n}}^{\infty} \frac{1}{(\gamma_n - \gamma_k)^2} + (\text{Gamma-factor derivatives}). \tag{8}$$

Since every term in the sum is positive, we have:

Lemma 2.3. $H_1(\gamma_n) = c_1(\gamma_n) > 0$ *for every simple zero. Moreover,*

$$H_1(\gamma_n) \geq \frac{1}{\Delta_L^2} + \frac{1}{\Delta_R^2}, \tag{9}$$

where $\Delta_L = \gamma_n - \gamma_{n-1}$ *and* $\Delta_R = \gamma_{n+1} - \gamma_n$*.*

Definition 2.4 (Hadamard amplitude)**.** The *Hadamard amplitude* of γ_n is

$$A(\gamma_n) = B(\gamma_n)^2 + 2\, H_1(\gamma_n),$$

where $B = \mathrm{Im}(c_0)$ and $H_1 = c_1$. We distinguish two variants:

- A_Λ: amplitude computed from the completed Λ-function (includes Gamma-factor contributions);
- A_L: amplitude computed from the primitive L-function (Gamma-factor terms omitted).

The relation is $c_m^\Lambda = c_m^L + \frac{1}{m!}\,(\gamma'/\gamma)^{(m)}(\rho_n)$.

3 Amplitude–Gap Anti-Correlation

3.1 Unconditional lower bound

Theorem 3.1 (Amplitude–gap lower bound)**.** *Let* $L(s) \in \mathcal{S}$ *have simple zeros* $\rho_n = \frac{1}{2} + i\gamma_n$ *on the critical line. Then*

$$A(\gamma_n) \;\geq\; \frac{2}{\Delta_{\min}(\gamma_n)^2}, \tag{10}$$

where $\Delta_{\min} = \min(\Delta_L,\, \Delta_R)$*,* $\Delta_L = \gamma_n - \gamma_{n-1}$*, and* $\Delta_R = \gamma_{n+1} - \gamma_n$*.*

Proof. Step 1. Since $B^2 \geq 0$, we have $A = B^2 + 2H_1 \geq 2H_1$.

Step 2. By lemma 2.3, $H_1 \geq 1/\Delta_L^2 + 1/\Delta_R^2$. Both terms are positive, so the sum exceeds its largest term:

$$\frac{1}{\Delta_L^2} + \frac{1}{\Delta_R^2} \;\geq\; \frac{1}{\Delta_{\min}^2}.$$

Step 3. Combining: $A \geq 2H_1 \geq 2/\Delta_{\min}^2$. □

Corollary 3.2. *If the gap distribution of $L(s)$ is nondegenerate (i.e., not all gaps are equal), then $\rho_{\mathrm{Sp}}(A, g_{\min}) < 0$.*

Proof. The lower bound $A \geq 2/\Delta_{\min}^2$ diverges as $\Delta_{\min} \to 0$, while A is bounded above for any finite collection of zeros (since H_1 is a convergent sum). Hence zeros with small g must have large A, and the rank ordering imposes a negative Spearman correlation. □

Remark 3.3 (Tightness)**.** The bound $A \geq 2/\Delta_{\min}^2$ is "almost tight" in the following numerical sense. For $\zeta(s)$ with $n = 910$ interior zeros ($\gamma \leq 2000$), the nearest-neighbour contribution $H_1^{\mathrm{NN}} = 1/\Delta_L^2 + 1/\Delta_R^2$ accounts for 66.4% of H_1, and $2H_1$ accounts for 86.7% of A. Thus the two-term lower bound captures the dominant mechanism. The fit $E[A \mid g] \approx a/g^2 + b$, where $g = g_{\min}^{\mathrm{GUE}}$ is the normalised gap, yields $a = 2.63$ (consistent with $a \geq 2\bar{d}^2 \approx 1.4$ from theorem 3.1, where $\bar{d}$ is the local density) with $R^2 = 0.987$.

3.2 Numerical verification for $\zeta(s)$

We verify theorem 3.1 on the Riemann zeta function using 1517 zeros with $\gamma \leq 2000$, of which the central 60% ($n = 910$) are used for unbiased gap estimation. The amplitude A_L is computed via the zero-sum method (Hadamard partial sums with all 1517 terms), while A_Λ is obtained by adding the Stirling-expanded Gamma contributions.

Statistic	**Value**	p**-value**	**Verdict**
$\rho(A_L,\ g_{\min})$	-0.7996	2.6×10^{-203}	✓
$\rho(A_\Lambda,\ g_{\min})$	-0.8998	$< 10^{-300}$	✓
$\rho(A_\Lambda,\ g_{\mathrm{right}})$	-0.4553	9.0×10^{-48}	✓
$2H_1/A$ (mean)	0.867	—	H_1 dominates
H_1^{NN}/H_1 (mean)	0.664	—	NN dominates
$E[A \mid g] \sim a/g^2 + b$	$R^2 = 0.987$	—	Bound tight

The 8-bin conditional expectation $E[A \mid g_{\min} \in \mathrm{bin}_j]$ is strictly monotone decreasing: $22.76 \to 14.20 \to 10.42 \to 8.17 \to 6.55 \to 5.46 \to 4.56 \to 4.02$ (all seven consecutive pairs satisfy $E[A]_j > E[A]_{j+1}$).

3.3 Conditional monotonicity under pair correlation

Proposition 3.4 (Conditional monotonicity)**.** *Under Montgomery's pair correlation conjecture, the conditional expectation satisfies*

$$E[A \mid g_{\min} = g] \;=\; \frac{2a}{g^2} + c + O(g) \qquad \text{as } g \to 0,$$

where $c = 2\int_1^\infty R_2(x)/x^2\,dx = 1.9527$ captures the tail contribution from distant zeros, and a is a constant depending on the pair correlation kernel.

Proof sketch. The nearest-neighbour contribution $H_1^{\mathrm{NN}} = 1/\Delta_L^2 + 1/\Delta_R^2$ dominates at small g, giving the $2a/g^2$ term (with a determined by the conditional distribution of Δ_L and Δ_R given their minimum). The tail contribution $H_1^{\mathrm{tail}} = \sum_{|k-n|\geq 2} 1/(\gamma_n - \gamma_k)^2$ converges to a height-independent constant under pair correlation universality:

$$E[H_1^{\mathrm{tail}}] = \bar{d}^2 \int_1^\infty \frac{R_2(x)}{x^2}\,dx + O(\bar{d}),$$

where $R_2(x) = 1 - (\sin \pi x/\pi x)^2$ is the GUE pair correlation function. Numerical evaluation gives $\int_1^\infty R_2(x)/x^2\,dx = 0.9764$, yielding $c = 2 \times 0.9764 = 1.9527$. □

4 Height Scaling of $A(\gamma)$

The bin-averaged amplitude follows a logarithmic-squared scaling:

$$\langle A(\gamma)\rangle_{\text{bin}} \approx 0.49\left(\log(\gamma/(2\pi))\right)^2 - 1.24\,\log(\gamma/(2\pi)) + 2.14, \tag{11}$$

with $R_w^2 = 0.975$ over 10 bins and 300 zeros of $\zeta(s)$ ($\gamma \in [14.1, 541.8]$).

The mechanism is the dominance of H_1: the mean density $\bar{d} = \log(\gamma/(2\pi))/(2\pi)$ implies $\langle H_1\rangle \sim \bar{d}^2 \times$ (pair corr. integral) $\sim (\log\gamma)^2/(2\pi)^2$, while B^2 contributes a subleading $(\log\gamma)^1$ term. The decomposition is $2H_1/A = 77.5\%$ (mean), $B^2/A = 22.5\%$.

Remark 4.1. The measured coefficient 0.129 for $\langle H_1\rangle \sim \alpha\,(\log\gamma/(2\pi))^2$ exceeds the naive estimate $1/12 = 0.083$ by a factor of 1.55. This discrepancy is expected: the Hadamard partial sum with finite K terms underestimates H_1; the EM-corrected full sum gives α closer to the theoretical value.

5 Degree-Independent Universality

We verify the amplitude–gap anti-correlation across twelve L-functions spanning degrees 1–6 of the Langlands hierarchy. For GL(1) and GL(2) L-functions, the Cauchy-contour amplitude A_Λ naturally pairs with g_{right}; for GL(3) through GL(6), the zero-sum amplitude A_L pairs with $g_{\min}$ (see section 6 for the explanation of this pairing).

Table 1: Amplitude–gap Spearman correlation across twelve L-functions. All correlations are negative ($p < 10^{-4}$ in every case).

L-function	d	N	n	Gap metric	ρ	p
$\zeta(s)$	1	1	198	g_{right}	-0.590	6.1×10^{-20}
$L(s,\chi_3)$	1	3	72	g_{right}	-0.573	1.4×10^{-7}
$L(s,\chi_4)$	1	4	80	g_{right}	-0.553	1.0×10^{-7}
$L(s,\chi_5)$	1	5	46	g_{right}	-0.633	2.3×10^{-6}
$L(s,E_{11\text{a}1})$	2	11	92	g_{right}	-0.569	3.3×10^{-9}
$L(s,E_{37\text{a}1})$	2	37	112	g_{right}	-0.550	3.4×10^{-10}
$L(s,\text{sym}^2E_{11})$	3	121	96	$g_{\min}$	-0.420	1.9×10^{-5}
$L(s,\text{sym}^2E_{37})$	3	1369	48	$g_{\min}$	-0.490	3.8×10^{-4}
$L(s,\text{sym}^3E_{11})$	4	1331	101	$g_{\min}$	-0.520	2.5×10^{-8}
$L(s,\text{sym}^3E_{37})$	4	50653	131	$g_{\min}$	-0.514	3.5×10^{-10}
$L(s,\text{sym}^4E_{11})$	5	14641	132	$g_{\min}$	-0.485	3.7×10^{-9}
$L(s,\text{sym}^5E_{11})$	6	161051	71	$g_{\min}$	-0.423	2.4×10^{-4}

All twelve L-functions exhibit negative amplitude–gap correlation with $p < 10^{-4}$. The correlation strength $\rho \in [-0.63, -0.42]$ is robust across degree ($d = 1$–6), conductor ($N = 1$–161051), and root number ($\varepsilon = \pm1$). Notably, the σ-uniqueness diagnostic saturates at $d \geq 4$ (becoming trivially PASS for all σ), while the A–gap anti-correlation retains full discriminative power—the two diagnostics probe orthogonal aspects of zero structure.

Observation 5.1 (Adjacent-pair correlation)**.** For every L-function in table 1, $\rho(A_n, A_{n+1}) \in [+0.28, +0.53]$ (consistently positive). The mechanism is that adjacent zeros share a nearest-neighbour term in their Hadamard sums: if $\Delta_{n,n+1}$ is small, both A_n and A_{n+1} receive a large $1/\Delta^2$ contribution.

5.1 Uniform metric comparison

The metrics in table 1 vary by degree: $A_\Lambda/g_{\text{right}}$ for $d = 1, 2$ versus $A_L/g_{\min}$ for $d \geq 3$. To enable direct cross-degree comparison, we re-measure all L-functions using the uniform pair $(A_L, g_{\min})$ with Spearman rank correlation (table 2).

Table 2: Method-matched amplitude–gap correlation across degrees 1–6. $\delta_{\text{arith}} = |\rho^{\text{ref}}_{\text{GUE}}| - |\rho_{\text{S}}|$ measures genuine arithmetic damping relative to the appropriate GUE baseline. $\dagger = T$-limited (actual δ may be smaller).

L-function	d	n	ρ_{S}	**Method**	**GUE ref.**	δ_{arith}	**Sig.**
$\zeta(s)$	1	$\sim$400	-0.929	$\pm$300, $T=2000$, trim 20%	-0.967 (local)	0.038	n.s.
$L(s, E_{11a1})$	2	$\sim$54	-0.658	full, $T=500$, trim 20%	-0.912 (full)	0.254	$\sim 2.4\sigma$
$L(s, E_{37a1})$	2	$\sim$66	-0.686	full, $T=500$, trim 20%	-0.912 (full)	0.226	$\sim 2.1\sigma$
$L(s, \text{sym}^2 E_{11})$	3	96	-0.422	full, $T \leq 35$	-0.912 (full)	0.490	T-lim†
$L(s, \text{sym}^2 E_{37})$	3	48	-0.492	full, $T \leq 35$	-0.912 (full)	0.420	T-lim†
$L(s, \text{sym}^3 E_{11})$	4	101	-0.520	full, $T \leq 35$	-0.912 (full)	0.392	T-lim†
$L(s, \text{sym}^3 E_{37})$	4	131	-0.514	full, $T \leq 35$	-0.912 (full)	0.398	T-lim†
$L(s, \text{sym}^4 E_{11})$	5	132	-0.485	full, $T \leq 35$	-0.912 (full)	0.427	T-lim†
$L(s, \text{sym}^5 E_{11})$	6	71	-0.423	full, $T \leq 35$	-0.912 (full)	0.489	T-lim†

Under method-matched conditions (table 2), $\zeta(s)$ achieves $\rho_{\text{S}} = -0.929$ ($T = 2000$, ± 300 local, 20% trim), GL(2) reaches -0.66 to -0.69 ($T = 500$, trim), while $d \geq 3$ remains at $\rho_{\text{S}} \in [-0.52, -0.42]$ with short T-ranges that preclude trim correction. The residual arithmetic damping δ_{arith} relative to the method-matched GUE baseline is 0.04 for $d = 1$ (not significant), 0.23–0.25 for $d = 2$ ($\sim 2\sigma$), and 0.39–0.49 for $d \geq 3$ (T-limited lower bounds).

Remark 5.2 (Locality and T-range effects)**.** The gap between $|\rho_{\text{GUE}}|$ and $|\rho_L|$ in table 2 ($\sim$0.38) is *not* a pure arithmetic effect. It conflates three distinct contributions: (i) *T-range truncation*: re-measuring $\zeta(s)$ with $T = 2000$ and 20% trim (versus $T = 600$, no trim) raises $|\rho_{\text{S}}|$ from 0.44 to 0.93; (ii) *locality of the sum*: restricting A to ± 300 nearest neighbours strengthens ρ_{S} in GUE from -0.912 (full sum) to -0.967 (± 300 local), since distant eigenvalues add noise rather than signal; and (iii) *genuine arithmetic damping*: under matched conditions (± 300 local, 20% trim), $\zeta(s)$ gives $\rho_{\text{S}} = -0.929$ versus GUE -0.967, a residual $\Delta\rho \approx 0.04$. For GL(2) ($T = 500$, trim), $\rho \approx -0.67$, giving $\Delta\rho \approx 0.30$—the damping is degree-dependent, not the universal 0.38 originally reported. The L-function values in table 2 are lower bounds limited by their short T-ranges; only $d = 1$ has been corrected to sufficient height.

Moreover, varying the window size W (number of nearest neighbours in the A-sum) reveals that the residual $\Delta\rho$ for $\zeta(s)$ is W-dependent: it decreases monotonically from 0.029 ($W = 5$) to 0.008 ($W = 750$), with Spearman trend $\rho_{\text{S}}(W, \delta) = -0.905$ ($p = 0.002$); for $W \geq 40$ the decrease is perfectly monotonic ($\rho_{\text{S}} = -1.0$). The decrease is driven by $|\rho_\zeta|$ strengthening from 0.95 to 0.96 as distant zeros contribute additional arithmetic correlation, while $|\rho_{\text{GUE}}| \approx 0.97$ is already saturated at $W = 5$. Fitting three candidate models to $\delta(W)$ over $W = 5$–750 ($N = 4{,}000$ zeros, bootstrap 95% CI), the exponential decay $\delta = a\, e^{-bW}$ ($a = 0.030 \pm 0.001$, $b = 0.0017 \pm 0.0003$) is overwhelmingly favoured ($\Delta\text{AIC} > 30$ versus both the power-law and offset models), predicting $\delta \to 0$ as $W \to \infty$. At $W = 1{,}500$ ($n_\zeta = 600$), $|\rho_\zeta| = 0.972$, matching $|\rho_{\text{GUE}}|$ to within 0.06%. Thus the *arithmetic nonlocality lag*—the slower spatial decay of inter-zero correlations in $\zeta(s)$ compared to GUE—is a finite-window artefact, and the $d = 1$ amplitude–gap correlation converges to full GUE universality under sufficient window size.

Remark 5.3 (Nearest-neighbour mechanism)**.** Decomposing A_{bare} into its nearest-neighbour contribution A_{NN} (the j=1 terms in S_1 and H_1) and the remainder A_{rest} reveals that the anti-correlation is driven by the nearest-neighbour term. For $\zeta(s)$ ($T = 2000$, 20% trim) across $W \in \{5, 20, 100, 500\}$: (i) A_{NN} accounts for 64–71% of A_{bare}; (ii) $|\rho_{\text{S}}(A_{\text{NN}}, g)| \geq |\rho_{\text{S}}(A_{\text{bare}}, g)|$ at

every W (e.g. -0.977 vs -0.971 at W=500); (iii) the simple predictor $2/g^2$ (pure H_1 nearest-neighbour) achieves $\rho = 0.97$ with A_{bare}. GUE matrices exhibit the same pattern at the same fractions (64%, $|\rho_{\text{NN}}|$=0.988). Since the nearest-neighbour H_1 term $1/(t_n - t_{n\pm1})^2$ is a monotonically decreasing function of the local gap g, the anti-correlation is a direct arithmetic consequence of the Hadamard product structure of $\xi(s)$, not a dynamical or statistical coincidence.

Remark 5.4 (Conductor independence within $d = 1$)**.** Cross-validation across GL(1) L-functions—$\zeta(s)$ ($q = 1$) and Dirichlet L-functions $L(s, \chi)$ with $q = 3, 4, 5$—under uniform conditions ($T = 500$, full sum, 20% trim) confirms that the amplitude–gap correlation is conductor-independent within degree 1: ρ_{S} ranges from -0.48 to -0.57, with maximum pairwise difference 0.09. This is within the 95% confidence interval width ~ 0.14 (estimated from $n \approx 215$–317). The damping factor is therefore controlled by the *automorphic degree* d, not by the conductor q.

5.2 Cross-family universality under canonical metric

The preceding analyses used mixed amplitude–gap pairings (table 1) or the uniform $(A_L, g_{\min})$ metric (table 2). We now present the strongest test of universality: nine L-functions measured under the *canonical* pairing $(A_\Lambda, g_{\min}^{\text{GUE}})$ identified in observation 6.1, with Gamma-corrected amplitudes and GUE-normalised gaps.

Method. For each L-function $L(s)$ of degree d, conductor N: (1) collect all zeros γ_n in $[0, T]$ via PARI/GP; (2) compute the Hadamard amplitude $A_\Lambda(\gamma_n)$ from the full zero-sum with Gamma corrections $c_0^\Lambda = c_0^L - \text{Im}(\Gamma'_\infty/\Gamma_\infty)$, $c_1^\Lambda = c_1^L + \text{Re}(\Gamma''_\infty/\Gamma_\infty)$; (3) normalise gaps by the theoretical density $\bar{d}(t) = (2\pi)^{-1}[\, d \log(t/(2\pi)) + \log N\,]$; (4) trim the outermost 20% on each side (central 60%); (5) report Spearman $\rho(A_\Lambda, g_{\min}^{\text{GUE}})$ with 95% confidence intervals.

Table 3: Canonical-metric amplitude–gap correlation across eleven L-functions (seven self-dual, four non-self-dual), spanning degrees 1–4. All p-values are below 10^{-76}.

L-function	d	N	n	ρ_{S}	95% **CI**	**Root #**
$\zeta(s)$	1	1	910	-0.900	$[-0.912, -0.887]$	$+1$
$L(s, \chi_{-3})$	1	3	494	-0.891	$[-0.908, -0.871]$	-1
$L(s, \chi_5^{(1)})$	1	5	238	-0.903	$[-0.929, -0.866]$	$\varepsilon \in S^1$
$\overline{L}(s, \chi_5^{(1)})$	1	5	238	-0.875	$[-0.910, -0.828]$	$\bar{\varepsilon} \in S^1$
$L(s, \chi_7^{(1)})$	1	7	254	-0.871	$[-0.908, -0.820]$	$\varepsilon \in S^1$
$\overline{L}(s, \chi_7^{(1)})$	1	7	255	-0.901	$[-0.927, -0.866]$	$\bar{\varepsilon} \in S^1$
$L(s, E_{11a1})$	2	11	437	-0.914	$[-0.928, -0.897]$	$+1$
$L(s, \Delta)$	2	1	324	-0.876	$[-0.899, -0.848]$	$+1$
$L(s, \text{sym}^2 E_{11})$	3	121	96	-0.885	$[-0.920, -0.830]$	$+1$
$L(s, \text{sym}^3 E_{11})$	4	1331	327	-0.905	$[-0.928, -0.873]$	$+1$
$L(s, \text{sym}^3 E_{19})$	4	6859	357	-0.908	$[-0.929, -0.881]$	$+1$
Self-dual mean ($n = 7$)				-0.896		$\varepsilon = \pm1$
Non-self-dual mean ($n = 4$)				-0.887		$\varepsilon \in S^1$
Overall mean ($n = 11$)				-0.894		
Overall std				0.014		
Overall range				0.043		

Observation 5.5 (Root-number independence)**.** The mean Spearman correlation is -0.896 for self-dual families ($\varepsilon = \pm1$, including two GL(4) entries) and -0.887 for non-self-dual families ($\varepsilon \in S^1$), a difference of 0.009 that is well within the individual 95% confidence intervals. The correlation is therefore independent of the root number, ruling out the hypothesis that the self-dual constraint $\varepsilon = \pm1$ is a necessary condition for the amplitude–gap anti-correlation.

For non-self-dual $L(s,\chi)$, the negative-imaginary-part zeros of $L(s,\chi)$ are the positive-imaginary-part zeros of $L(s,\bar{\chi})$, so the "conjugate sum" in the Hadamard decomposition $\sum_k 1/(\gamma_n + \tilde{\gamma}_k)$ uses zeros of $\bar{\chi}$ rather than χ itself. Numerically, the "proper" (cross-character) and "naïve" (self-paired) amplitudes differ by $|\Delta\rho| < 0.0003$, confirming that the conjugate sum contributes a smooth, position-insensitive baseline at $O(n(T)/(2T))$.

Observation 5.6 (Conjugate-pair consistency)**.** For each conjugate pair $(\chi,\bar{\chi})$: $|\rho(\chi) - \rho(\bar{\chi})| = 0.029 \pmod 5$ and $0.031 \pmod 7$, with alternating sign. This is consistent with statistical fluctuation from independent zero sets of equal density, providing an internal consistency check.

Observation 5.7 (Degree-dependent $2H_1/A_L$ ratio)**.** The ratio $2H_1/A_L$ (computed under the uniform metric) is: 0.666 (GL(1)) $\to$ 0.652–0.675 (GL(2)) $\to$ 0.678–0.748 (GL(3)) $\to$ 0.661–0.676 (GL(4)) $\to$ 0.661 (GL(5)) $\to$ 0.657 (GL(6)). All values except the GL(3) outlier 0.748 cluster near $2/3 \approx 0.667$. This 2/3 value matches the GUE nearest-neighbour dominance ratio $H_1^{\mathrm{NN}}/H_1 = 0.669$ (section 7.1), suggesting that the nearest-neighbour contribution to H_1 controls the $2H_1/A_L$ ratio universally.

6 A_Λ versus A_L Dichotomy

The cross-verification experiment on $\zeta(s)$ reveals a fundamental dichotomy in the amplitude–gap pairing:

- A_Λ (completed, including Gamma): correlates with g_{right} ($\rho = -0.72$ at low t, decaying to -0.12 at $t \approx 1600$);
- A_L (primitive, no Gamma): correlates with $g_{\min}$ ($\rho = -0.80$ overall, mildly height-dependent).

The relation is:

$$c_0^\Lambda = c_0^L + \frac{1}{2}\,\psi\Big(\frac{\rho}{2}\Big) + \mathrm{const}, \tag{12}$$

where $\psi = \Gamma'/\Gamma$. The Gamma-factor shift $\sim \log(t/2\pi)$ grows with height, introducing a systematic directional bias that favours g_{right} at low t but dilutes at high t.

6.1 Height stability: $g_{\min}$ as canonical metric

To resolve boundary B-46 (height dependence of the amplitude–gap correlation), we partition the 910 interior zeros of $\zeta(s)$ into eight height bins and measure ρ in each:

Bin	$\bar{t}$	$\rho(A_\Lambda, g_{\min})$	$\rho(A_\Lambda, g_{\mathrm{right}})$	$\rho(A_L, g_{\min})$	$\rho(A_L, g_{\mathrm{right}})$	n
1	626	−0.814	−0.686	−0.956	−0.450	114
2	778	−0.889	−0.641	−0.973	−0.376	114
3	924	−0.938	−0.624	−0.963	−0.300	113
4	1064	−0.961	−0.504	−0.934	−0.216	114
5	1202	−0.977	−0.496	−0.906	−0.191	114
6	1337	−0.948	−0.339	−0.829	−0.057	113
7	1469	−0.954	−0.262	−0.823	+0.002	114
8	1599	−0.867	−0.120	−0.673	+0.159	114
Span		0.163	0.567	0.300	0.609	
Kendall τ		−0.29	+1.00	+0.86	+1.00	
p-value		0.40	5×10^{-5}	0.002	5×10^{-5}	

Observation 6.1 ($g_{\min}$ height stability)**.** The pairing (A_Λ, $g_{\min}$) is height-stable: Kendall $\tau = -0.29$ ($p = 0.40$, not significant), span $= 0.163 < 0.20$. All other three pairings show significant monotonic trends ($p < 0.01$). This identifies $g_{\min}$ as the *canonical gap metric* for the amplitude anti-correlation.

The mechanism is transparent: the lower bound $A \geq 2/\Delta_{\min}^2$ (theorem 3.1) is purely algebraic and height-independent. Since the bound is tight ($2H_1^{\mathrm{NN}}/A \approx 87\%$), the rank correlation is mechanically driven by the bound and inherits its height-independence. The g_{right} metric, by contrast, receives Gamma-factor contamination ($\sim \log(t)$ additive shift) that degrades monotonically with height.

Remark 6.2 (Cauchy-method verification)**.** An independent Cauchy-contour computation of A_Λ at $T \leq 300$ ($n = 136$, radius $= 0.01$, 128 quadrature points, `dps`$= 60$) yields $\rho(A_\Lambda, g_{\mathrm{right}}) = -0.580$ ($p = 1.4\times10^{-13}$), reproducing the original measurement -0.590 within $|\Delta\rho| = 0.010$. The discrepancy between this low-t value (-0.58) and the full-range value (-0.46) is fully explained by the height decay documented above.

7 GUE Prediction

To quantify the random-matrix content of the amplitude–gap correlation, we simulate 1000 realisations of 300×300 GUE matrices, extract 88,000 bulk eigenvalues (central 30%), and compute the Hadamard amplitude $A = \sum_{k\neq n} 1/(E_n - E_k)^2$ (with convergence at $K \geq 5$ nearest neighbours).

Quantity	**GUE prediction**	**Observed**	**Match**
$\rho(A, g_{\mathrm{right}})$	-0.499 ± 0.001	-0.578	86%
$\rho(A_n, A_{n+1})$	$+0.363$	$+0.28$–$+0.42$	Exact
NN contribution $2H_1^{\mathrm{NN}}/A$	68%	78%	Directional

The GUE reproduces 86% of the observed g_{right} correlation. The residual $\Delta\rho \approx 0.08$ has two candidate origins:

1. **Gamma-factor cross-term.** The Gamma contribution $\Delta A_\Gamma = -2S_1^L \cdot \mathrm{Im}(\psi(\rho/2)/2)$ introduces a height-dependent shift that strengthens the correlation at low t. Measured directly: $\rho(\Delta A_\Gamma, g_{\mathrm{right}}) = -0.70$ ($p \approx 0$), confirming that the cross-term is the primary source of the 14% residual.

2. **Arithmetic content.** The S_1^2/A ratio differs between GUE bulk (11%) and L-functions ($\sim$23%), reflecting the non-cancellation of arithmetic coefficients in the alternating sum B.

7.1 Full Laurent amplitude and N-independence

The preceding analysis used a nearest-neighbour approximation $A \approx 2H_1$. We now compute the *full* Laurent amplitude $A = S_1^2 + 2H_1$ for GUE(N) matrices with $N \in \{50, 100, 200, 500, 1000, 2000\}$, reporting Spearman correlations with $g_{\min}^{\mathrm{GUE}}$ (central 80% of eigenvalues, 20–100 realisations per N). We also measure the effect of restricting A to a local window of ±300 nearest eigenvalues, matching the methodology used for $\zeta(s)$ in section 5.1.

N	$\rho_{\mathrm{S}}^{\mathrm{full}}$	$\rho_{\mathrm{S}}^{\mathrm{local}}$	$\lvert\Delta\rho\rvert$	$2H_1/A$	H_1^{NN}/H_1
200	-0.915	-0.915	0.000	0.779	0.672
500	-0.913	-0.928	0.015	0.779	0.669
1000	-0.911	-0.963	0.052	0.778	0.669
2000	-0.912	-0.967	0.054	0.778	0.669

Four conclusions follow.

1. **Spearman stability.** $\rho_{\mathrm{S}}^{\mathrm{full}}(A, g_{\min}) = -0.912 \pm 0.002$ is constant across $N = 200$–2000 ($p \approx 0$ for all N). The amplitude–gap anti-correlation is an intrinsic GUE property, independent of matrix size.

2. **Locality enhancement.** Restricting A to ± 300 nearest eigenvalues *strengthens* the correlation from -0.912 to -0.967 at $N = 2000$ ($\Delta\rho = +0.054$). Distant eigenvalues contribute noise to S_1 without improving gap prediction. This effect saturates by $N \approx 1000$.

3. **Arithmetic attenuation—resolved for $d{=}1$.** Under matched conditions (± 300 local, 20% trim), GUE gives $\rho_{\mathrm{S}} = -0.967$ while $\zeta(s)$ gives -0.929 (remark 5.2). This residual $\Delta\rho \approx 0.04$ vanishes exponentially as W grows ($\delta \propto e^{-0.0017W}$, $\Delta\mathrm{AIC} > 30$); at $W = 1{,}500$, $|\rho_\zeta| = 0.972 \approx |\rho_{\mathrm{GUE}}|$. For $d = 1$ the amplitude–gap correlation thus achieves full GUE universality; the earlier naïve gap 0.38 conflated finite-T, finite-W, and methodology-mismatch effects.

4. **Pearson artefact.** Pearson ρ_{P} fluctuates between -0.12 and -0.33 because the heavy-tailed distribution of A violates the linearity assumption. Spearman, being rank-based, is immune. We therefore report only Spearman throughout.

The ratio H_1^{NN}/H_1 converges to 2/3 (reaching 0.669 at $N = 2000$), matching the L-function observation. In contrast, $2H_1/A = 0.78$ (GUE) differs from 0.66 (L-functions, $d \geq 4$): the overall ratio is *not* a GUE universal, though the nearest-neighbour dominance fraction is.

8 Partial Correlation

To disentangle the "shared gap" effect from the intrinsic amplitude–amplitude coupling, we compute the partial correlation $\rho(A_n, A_{n+1} \mid g_{n,n+1})$, controlling for the gap between the n-th and $(n{+}1)$-th zeros.

For $\zeta(s)$ ($n = 910$): $\rho_{\mathrm{partial}}(A_n, A_{n+1} \mid g_{n,n+1}) = +0.391$ ($p = 1.3 \times 10^{-18}$). Even after removing the shared-gap effect, adjacent amplitudes remain positively correlated. The mechanism is the shared nearest-neighbour term: if γ_n and γ_{n+1} are both close to γ_{n+2}, both amplitudes receive large H_1 contributions from γ_{n+2}.

9 σ–A Cross-Check

The σ-profile expansion (Corollary 4.3 of [4]) provides an independent route to $A(\gamma)$: fitting the curvature profile $\kappa(\sigma, \gamma) = 1/(\sigma - \frac{1}{2})^2 + A + O((\sigma - \frac{1}{2})^2)$ yields a coefficient $c(\gamma)$ that should equal $A(\gamma)$ from the Laurent expansion.

For $\zeta(s)$ with 20 zeros: $\rho(\sigma\text{-scan } c(\gamma), \text{ Laurent } A(\gamma)) = 0.99999744$, with maximum relative discrepancy 0.19%. This six-digit agreement between two completely independent computational methods—one from σ-direction fitting, the other from Cauchy contour integration—validates the amplitude definition and rules out systematic numerical artefacts.

10 Discussion

10.1 Summary of results

We have established that the Hadamard amplitude $A(\gamma)$ is a meaningful invariant of individual zeros of automorphic L-functions:

(i) The unconditional bound $A \geq 2/\Delta_{\min}^2$ (theorem 3.1) is a direct consequence of Hadamard's product formula and requires no unproven hypotheses.

(ii) The anti-correlation is verified across twelve L-functions of degree 1–6, establishing degree-independent universality. Under the canonical $(A_\Lambda, g_{\min}^{\mathrm{GUE}})$ metric, eleven families of degree 1–4 give $\rho = -0.894 \pm 0.014$ (table 3), with both self-dual and non-self-dual root numbers represented.

(iii) The GUE random matrix ensemble produces a Spearman anti-correlation $\rho_{\mathrm{S}}^{\mathrm{full}} = -0.912$ (full sum, 20% trim, constant over $N = 200$–2000), strengthening to $\rho_{\mathrm{S}}^{\mathrm{local}} = -0.967$ when restricted to ± 300 nearest eigenvalues. Under matched conditions, $\zeta(s)$ gives -0.929: arithmetic weakens the correlation by $\Delta\rho \approx 0.04$, with larger damping ($\Delta\rho \approx 0.30$) at $d = 2$.

(iv) The pairing $(A_\Lambda, g_{\min})$ is height-stable, while $(A_\Lambda, g_{\mathrm{right}})$ decays, identifying $g_{\min}$ as the canonical metric.

10.2 Physical interpretation

The amplitude $A(\gamma)$ can be interpreted as the *spectral influence radius* of a zero: zeros with large A exert stronger influence on the local analytic structure, and theorem 3.1 shows that this influence is inversely proportional to the square of the nearest-neighbour distance. The analogy with the Berry–Keating spectral interpretation $A \sim$ local density of states of a quantised Hamiltonian is suggestive but remains unproven.

10.3 Open problems

1. **Analytic derivation of $\rho \approx -0.47$.** Under the uniform $(A_L, g_{\min})$ metric, all nine L-functions give $\rho_{\mathrm{S}} \in [-0.525, -0.422]$ (table 2). Can this specific range be derived from the pair correlation function and the GUE value -0.857?

2. **Mechanism of degree-dependent damping.** For $d = 1$, the residual $\Delta\rho \approx 0.04$ at $W = 300$ vanishes exponentially as W grows (remark 5.2), confirming that $\zeta(s)$ converges to full GUE universality; the damping is a finite-window artefact. For GL(2) ($\Delta\rho \approx 0.30$, T-limited), the question remains: is the large damping at degree $d \geq 2$ also a finite-T/finite-W effect, or is there an intrinsic arithmetic attenuation traceable to the Gamma factor complexity or the Euler product structure? Remark 5.4 rules out conductor dependence within $d = 1$, pointing toward a purely degree-driven mechanism.

3. $2H_1/A_L \approx 2/3$ **universality.** The ratio $2H_1/A_L$ clusters near 2/3 across all degrees (observation 5.7), matching the GUE nearest-neighbour dominance $H_1^{\mathrm{NN}}/H_1 = 0.669$. Is the equality $2H_1/A_L = 2/3$ exact in any limit?

4. **σ-localisation and A–gap duality.** The σ-profile $\kappa(\sigma) = 1/\varepsilon^2 + A + \cdots$ and the gap bound $A \geq 2/\Delta_{\min}^2$ suggest a duality between the transverse (σ-direction) and longitudinal (t-direction) structure of zeros.

5. **GL(7) and beyond.** The universality now extends through sym^5 (degree 6); extension to degree 7 and higher would further test the Langlands hierarchy. The main obstacle is computational: `lfunzeros` at degree 6 already requires $\sim$400 s per L-function.

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